MILITARY SPECIF1CATION

MALF 3 BUFFER LOGIC: 11738803

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers one type of electronic assembly known as MALF 3 Buffer Logic: 11738803 (see 6.1).

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

Military

MIL-F-13926	Fire Control Materiel: General specification Governing the Manufacture and inspection of
MIL-I-45607	Inspection Equipment; Acquisition Maintenance, and Disposition of
MIL-S-45743	Soldering, Manual Type, High Reliability Electrical, Electronic, Instrument, Communication, and Radar for Aerospace and Control Systems, Procedures for

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MIL-S-46844	Solder Bath Soldering of Printed Wiring Assemblies, Automatic Machine Type

STANDARDS

Military

MIL-STD-105	Sampling Procedures and Tables for inspection by Attributes
MIL-STD-109	Quality Assurance Terms and Definitions
MIL-STD-810	Environmental Test Methods

DRAWINGS

U. S. Army, Frankford Arsenal

F11738803	MALF 3 Buffer Logic: 11738803

PACKAGING DATA Sheet

11738803	Packaging of MALF 3 Buffer Logic: 11738803

(Copies of specifications. standards, drawings, and packaging data sheets required by suppliers in Connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

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3. REQUIREMENTS

3.1 Fabrication. The MALF 3 Buffer Logic, herein referred to as the assembly, shall be manufactured in accordance with Drawing F11738803 and drawings pertaining thereto.

3,2 General specification. The assembly shall meet the following requirements, where applicable, of MIL-F-13926:

a. Order of precedence

b. Dimensions and tolerances

c. Part identification and marking

d. Electrical and electronic assemblies

e. Part identification and marking

f. Workmanship.

3. 2.1 Soldering. Soldering shall be in accordance with MIL-S-45743 or MIL-S-46844, except that bare copper wire exposure shall he perrnitted at clipped ends of component leads.

3.3 Performance. Unless otherwise specified, the assembly Shall meet the performance requirements of this specification at standard ambient temperatures between 60 degrees Fahrenheit (°F) and 90° F.

3. 3.1 Loads, power, and signals. The assembly shall perform as specified herein when the loads, power, and signals of table I are applied as specified.

3. 3.1.1 Reset signal. With the type-B signal (item 3, 3 of table 1) applied to P1-10 as shown in figure 1, the output at P1-29 and P1-30 shall be as shown in figure 1. (These signals may be recurring.)

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TABLE I
LOADS, POWER, AND SIGNALS

Item	Conditions	Characteristics	Connections
1.	Loads		Connected between the following pins:
1.1	Impedance	160Ω ±5 percent in series with 62 pf ±10 percent	P1-30 and P1-29
1.2	Impedance	160Ω ±5 percent in series with 62 pf ±10 percent	P1-16 and P1-17
1.3	Impedance	160Ω ±5 percent in series with 62 pf ±10 percent	P1-19 and P1-18
1.4	NAND Gate	1 Gate-DTμL 9946 or equivalent	P1-3 and P1-7
1.5	NAND Gate	2 Gates-DTμL 9946 or equivalent	P1-32 and P1-7
2.	Power Source		
2.1	+5 Vdc	Tolerance: ±0.2 Vdc Ripple: ≤25 mV p-p	P1-5(+) and P1-7 (-)
3.	Signal Sources		Applies as specified herein.
3.1	Digital Type-A	Logical one: +4.0 ±1.0V Source impedance: 6KΩ ±15 percent Logical zero: +0.2 ±0.2V Pulse width and PRF as specified herein.	
3.2	Digital Type-A1	Logical one: +4.0 ±1.0V Source impedance: 680Ω ±30 percent Logical zero: +0.2 ±0.2V Pulse width and PRF as specified herein.	
3.3	Digital Type-B	Logical one: +4.0 ±1.0V Source impedance: 150Ω±30 percent Logical zero: +0.2 ±0.2V Pulse width and PRF as specified herein.	

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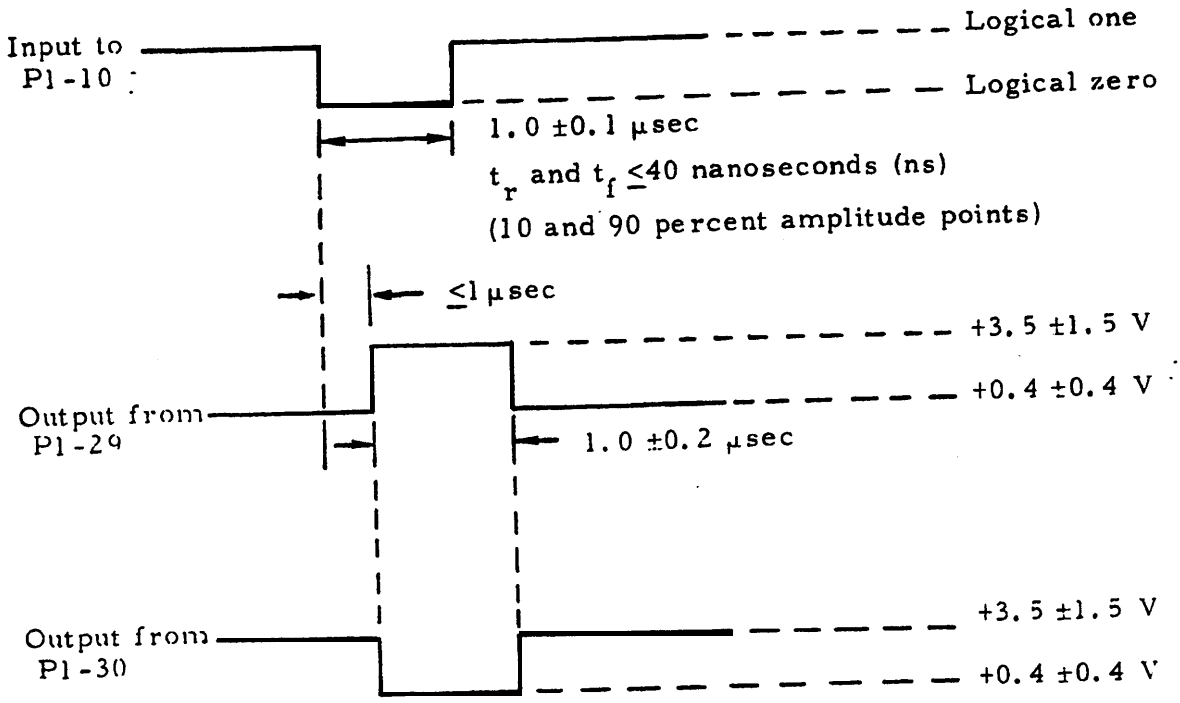


Figure- 1 Reset Signal Wave forms

3.3.1.2 Range signal. With no input signal applied to P1-1, the output at P1 -3 shall be +5.0 ± 1.0 Vdc. With P1-1 connected to P1-7, the output at P1 - 3 shall be +0.2 ± 0.2 V. With P1-1 disconnected from P1-7 and and P1-2 connected to P1-7,the output at P1-3 shall be +0.2 ±0.2V.

3. 3.1.3 Test range. With the type-B signals shown in fugure 2 applied to P1-23 and P1-24, the output at P1-6 shall be as shown in figure 2. (This signal may be recurring.)

3.3.1.4 A-trigger signal. With the type-A signal shown in figure 3 applied to P1-35, the outputs at P1-17 and P1-16 shall be as shown in figure 3. (The signal may be recurring.)

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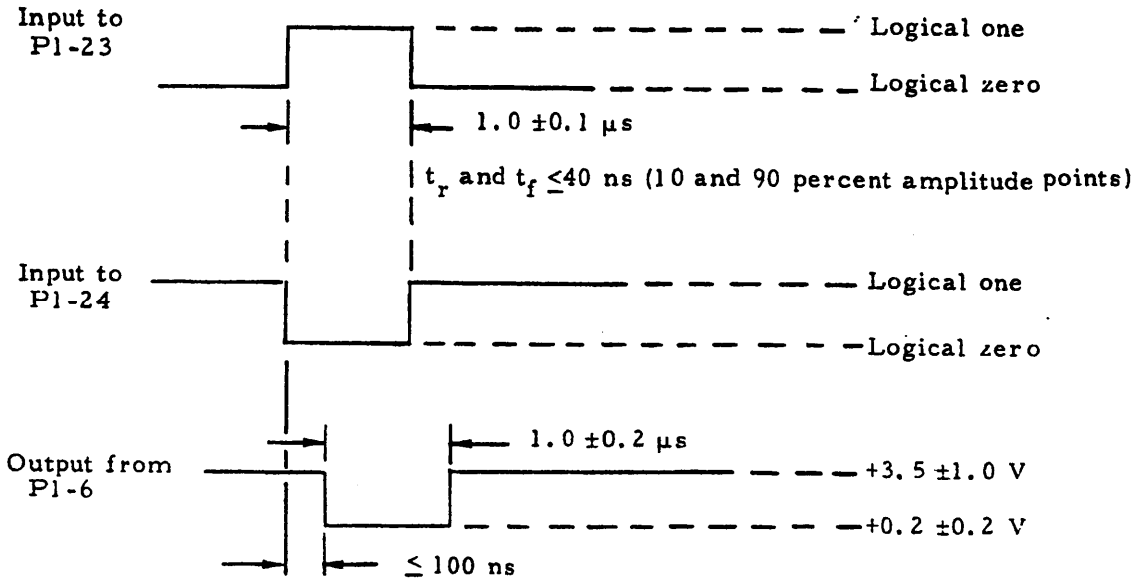


Figure 2. Test Range Input/Output Signal Waveforms

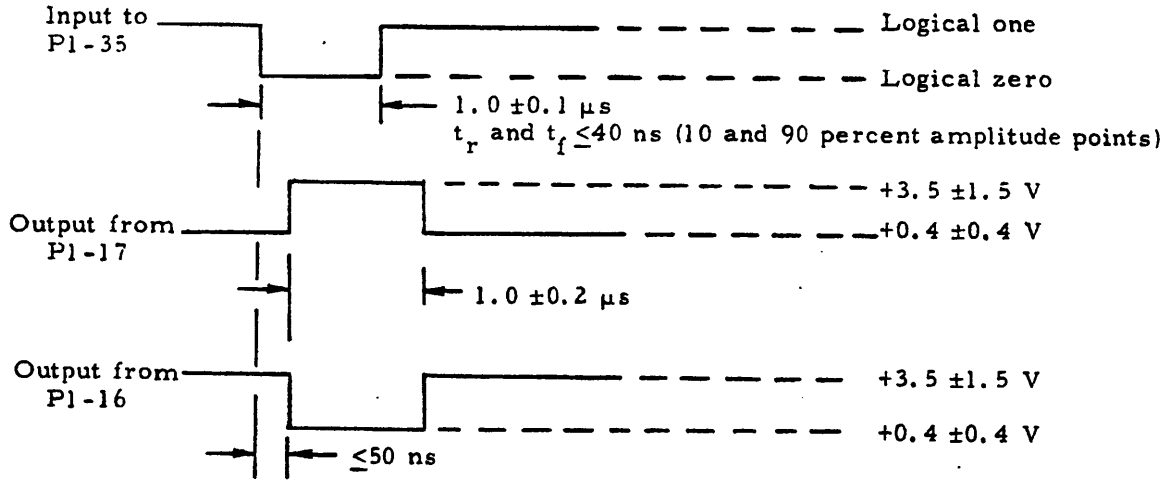


Figure 3. $\overline{\text{A-trigger}}$ Input/Output Signal Waveforms

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3,3.1.5 Video signal. With the type -A 1 signal shown in figure 4 applied to P1-37, the outputs at Pl-19 *and P1-18 shall be as shown on* figure 4. (These signals may be recurring.)

3. 3.1.6 MALF 3 signal.

3. 3.1.6.1 With Logical one type-A and type-A1 signals of table 1 applied to P1-35 and P1-37, respectively, and a momentary logical zero type-A signal applied to P1- 14, the output at P1-32 shall be +0.2±0.2V. With a logical zero type-A signal momentarily applied to P1-33, the output at P1-32 shall be +3.5 ±1.0V.

3. 3.1.6.2 With the logical zero type-A signal of table I applied momentarily to P1- 14, the output at P1-32 shall be +0. 2 ± 0.2V. With a type-A signal applied to P1 -35 and a type-Al signal applied to P1-37 with the waveforms shown on figure 5 (these signals may be recurring), the output at P1-32 shall be -3. 5 ± 1.0V.

3. 3.1.7 Continuity. With no signals or power applied, direct continuity (resistance equal to or less than 0.3 ohm), shall exist between pins P1-7 and P1-12, P1-11 and P1-30, P1-29 and P1-31, and between P1-29 and P1-13.

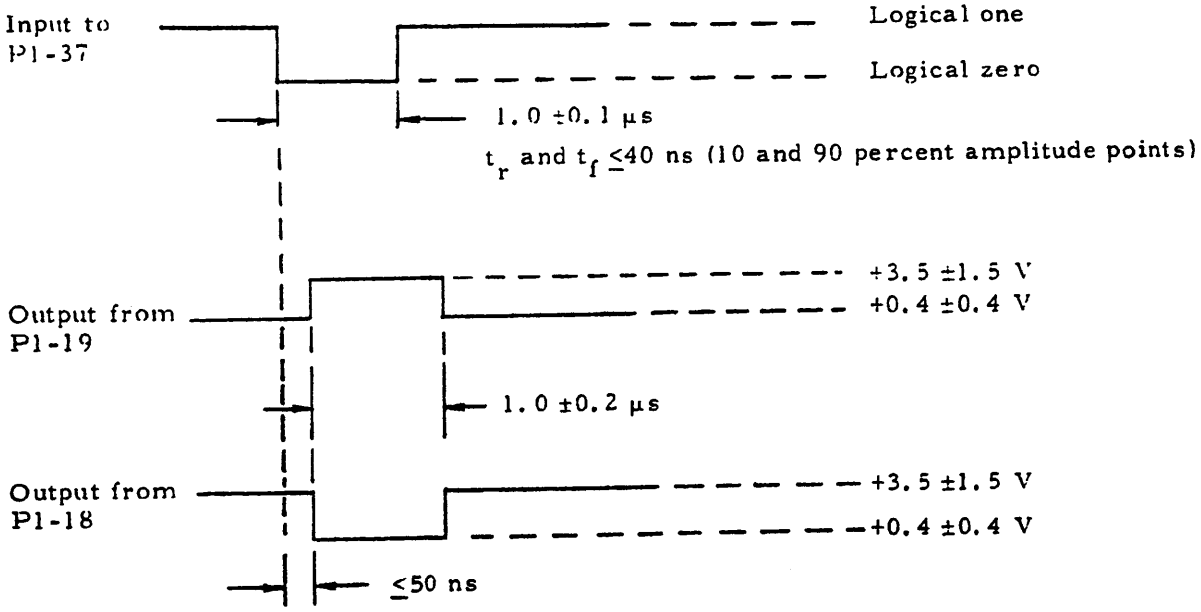
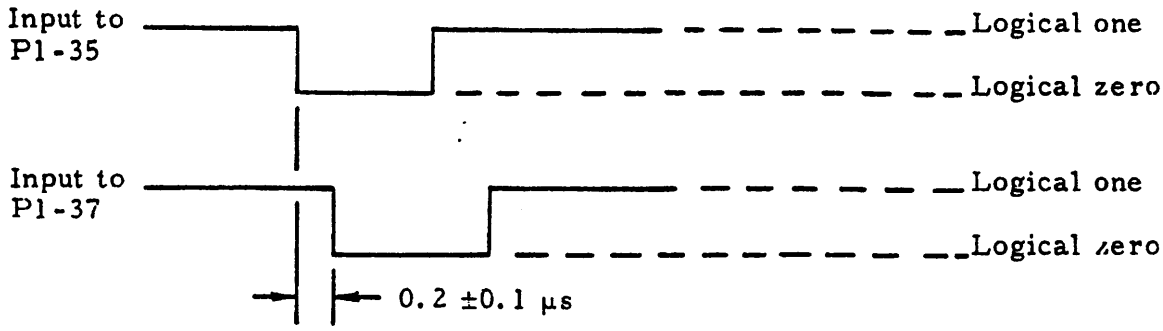


Figure 4. Video Input/Output Signal Waveforms

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Pulse widths = 1.0 ± 0.2µs

t_r and $t_f \leq$ 40 ns (10 and 90 percent amplitude points)

Figure 5. MALF 3 input Signal Waveforms

3.3.2 Environmental

3.3.2.1 Storage temperature. The assembly shall meet the requirements of 3.3.1 at ambient temperature after exposure to and thermal stabilization at -80°F and +160°F.

3.3.2.2 Operating temperatures. The assembly shall meet the requirements of 3.3.1 while exposed to and thermally stabilized at -40° F and +125°F, subsequent to which the requirements of 3.3.1 shall be met at ambient temperature.

3.3.2.3 Vibration, The assembly shall show no physical failure and shall meet the requirements of 3.3.1 after being exposed to the following vibratory conditions:

a.	Amplitude:	1/64 inch, constant (1/32 inch double amplitude)
b.	Vibratory motion:	Simple harmonic
c.	Sweep time:	5 to 55 to 5 Hertz in a five-minute period
d.	Vibration cycling:	Three linear or logarithmic sweeps applied to each of the three mutually perpendicular axes.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize-his own or any other-facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements

4.1.1 General provisions, The component and subassembly inspection requirements of MIL-F-13926 form a part of the quality assurance provisions of this specification. Definitions of inspection terms shall be as listed in MlL-STD-109.

4.2 First article [initial production) approval. The requirements for first article approval and the responsibility (Government or contractor) for first article testing shall be as specified in the contract The sample for first article approval tests shall consist of three assemblies. The sample shall be manufactured in the same manner, using the same materials, equipment, processes, and procedures as used in regular production. All parts and materials, including packaging and packing, shall be obtained from the same source of supply as used in regular production.

4.2.1 Government testing. When the Government is responsible for conducting first article approval tests, the contractor prior to submitting the sample to the Government, shall inspect the sample to insure conformance to to the requirements of this specification. A record of this inspection, including certificates of conformance for materials, shall be submitted with the sample.

4. 2.2 Contractor testing. When the contractor is responsible for conducting first article approval tests, the sample shall be inspected by the contractor for all the requirements of this specification, The sample and a record of this inspection, including certificates of conformance for materials, shall be submitted to the Government for approval. The Government reserves the right to witness the contractor's inspection.

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4.3 Inspection provisions.

4.3.1 Submission of product. Unless otherwise specified herein, or by the contracting officer, inspection lot size, lot formation and presentation of lots shall be in accordance with inspection and sampling procedures herein and submission of product provisions of MIL-STD-105.

4.3.2 Examination and tests.

4.3.2.1 Components and subassemblies. All components and subassemblies shall be inspected in accordance with the inspection provisions contained in the respective specifications and Supplementary Quality Assurance provisions (SCAP) Listed in the technical data package (TDP). In the absence of SQAP'S the applicable quality assurance provisions of MIL-F-13926 shall apply.

4.3.2, 2 Functional tests. The requirements and tests in table II shall be inspected on a 100 percent basis. All examinations and tests shall be conducted at the standard ambient temperature.

TABLE II

FUNCTIONAL TESTS

Classification	Characteristics	Requirements	Test Paragraph
101	$\overline{\text{Reset}}$ signal	3.3.1.1	4.6.2
102	$\overline{\text{Range}}$ signal	3.3.1.2	4.6.3
103	Test range	3.3.1.3	4.6.4
104	$\overline{\text{A-trigger}}$ signal	3.3.1.4	4.6.5
105	$\overline{\text{Video}}$ signal	3.3.1.5	4.6.6
106	$\overline{\text{MALF 3}}$ signal	3.3.1.6	4.6.7
107	Continuity	3.3.1.7	4.6.8

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4.4 Special sampling. The selection of samples shall be in accordance with section 7 of MIL-STD-105.

4.4.1 General. One assembly shall be selected at random by a Government representative as a control sample from each 50 produced. The control sample shall meet the requirements and tests in table III.

4.4.2 Environmental. Three assemblies shall be selected at random by a Government representative as a control sample from each 50 produced or from each month's production whichever occurs first. The control samples shall meet the requirements and tests in table IV.

4.4.3 Acceptance and rejection. Rejected lots shall be screened for all defective characteristics. Removal or correction of defective units and resubmittance of rejected lots shall be in accordance with the acceptance and rejection criteria specified in MIL-STD-105.

TABLE III

SPECIAL SAMPLING - GENERAL

Classification	Control Tests	Requirements	Test Paragraph
301	Fabrication	3.1	Applicable drawings- visual
302	General Specification	3.2	MIL-F-13926 visual

TABLE IV

SPECIAL SAMPLING - ENVIRONMENTAL

Classification	Control Tests	Requirements	Test Paragraph
303	Storage temperature	3.3.2.1	4.6.9.1
304	Operating temperature	3.3.2.2	4.6.9.1
305	Vibration	3.3.2.3	4.6.9.2

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4.5 Inspection equipment. Except as otherwise provided for by the contract, the contractor shall supply and maintain inspection equipment in accordance with the applicable requirements of MIL-I-45607. The inspection equipment listed in table V shall be utilized in performing the tests specified herein.

4.5.1 Government furnished inspection equipment. Where the contract provides for Government furnished inspection equipment, supply and maintenance of inspection equipment shall be in accordance with. the applicable requirements of MIL-I-45607.

4.5.2 Contractor furnished inspection equipment.

4.5.2.1 Government design. Unless otherwise specified in the contract, all inspection equipment specified by drawing number in specifications or SQAPs forming a part of the contract shall be supplied by the contractor in accordance with applicable technical data listed in the TDP.

4.5.2.2 Contractor design. The contractor shall design and supply inspection equipment compatible with the test methods and procedures specified in 4.6 of this specification and with the component inspection procedures specified in examination and test facilities provisions of MIL-F-13926. Since tolerance of inspection equipment is normally considered to be within 10 percent of the product tolerance for which intended, this inherent error in the inspection equipment design must be considered as part of the prescribed product tolerance limit. Thus, concept, construction, materials, dimensions, and tolerances used in the design of inspection equipment shall be so selected and controlled as to insure that the inspection equipment will reliably indicate acceptability of a product which does not exceed 90 percent of the prescribed tolerance limit and permit positive rejection when non-conforming. Construction shall be such as to facilitate routine calibration of inspection equipment.

4.6 Test methods and procedures.

4. 6.1 Test conditions. Unless otherwise specified, the test conditions shall be in accordance with the test facilities provisions of MIL-F-13926.

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TABLE V
INSPECTION EQUIPMENT

Item	Nomenclature	Description
1	DC Power Supply	+5.0 ±0.2V, Ripple: ≤25 mV p-p
2	DC Voltmeter	Range: 0 to 20V, Accuracy: ±1 percent
3	Signal source	Capable of providing dc levels and wave forms specified in item 3.1 of table I.

4.6.1.1 Initial conditions. Connect the loads and power to the assembly as specified in table I.

NOTE: The MALF 3 buffer logic may be performance tested using the digital logic type truth-table shown in table VI when tested at the system level,

4. 6.2 Reset signal test. Apply the digital type-B signal of table 1 to P1-10 as shown in figure 1. Measure the outputs at P1-29 and P1-30 to determine compliance with 3.3.1.1.

4. 6.3 Range signal test. With no input applied to P1-1, measure the output at P1-3 to determine compliance with 3.3.1.2. Connect P1-1 to P1-7. Measure the output at P1-3 to determine compliance with 3.3.1.2. Disconnect P1-1 from P1-7 and connect P1-2 to P1-7. Measure the output at P1-3 to determine compliance with 3.3.1.2.

4.6.4 Test range test. Apply the type -B input signal of figure 2 to P1-23 and P1-24. Observe the output at P1-6 to determine compliance with 3.3.1.3.

4.6.5 A-trigger signal test. Apply the type-A signal of figure 3 to P1-35. Observe the outputs at P1-17 and P1-16 to determine compliance with 3.3.1.4

4.6.6 Video-signal test. Apply the type-A1 input signal of figure 4 to P1-37. Observe the output at P1-19 and P1-18 to determine compliance with 3. 3. 1. 5

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4.6.7 MALF signal test

4.6.7.1 Apply the logical one type-A and type-A1 input signals (items 3.1 and 3.2 of table. I) to P1-35 and P1-37, respectively, and a momentary logical zero type -A signal to P1-14. Measure the output at P1-32 to determine compliance with 3.3.1.6.1. Then apply a momentary logical zero type-A signal to P1 -33 and, again, measure the output at P1-32 to determine compliance with 3.3.1.6.1.

4.6.7.2 Apply a logical zero type-A signal momentarily to P1-14. Measure the output at P1-32 to determine compliance with 3.3.1.6.2. Then apply type-A and type-Al signals as specified on figure 5 to P1-35 and P1-37, respectively. Measure the output at P1-32 to determine compliance with 3.3.1.6.2.

4.6.8 Continuity test, With no power or signals applied, measure the resistance between pins P1-7 and P1-12, P1-11 and P1 -30, P1-29 and P1-31, and P1-29 and P1-13 to determine compliance with 3.3.1.7.

4.6.9 Environmental tests.

4.6.9.1 Temperature. The test apparatus shall be in accordance with the applicabl test facilities provisions of MIL-F- 13926 except that the assembly may be mounted on the inside surface of the test chamber. Perform this test in accordance with Procedure I of MIL-F-13926, with conditions C (-40°F) and E (+125°F), and with the rate of temperature change during cycling not greater than 8°F per minute. At each of the specified operating temperatures, perform the tests of 4. 6.2 through 4.6.8 to determine compliance with 3.3.2.1 and 3.3.2.2.

4.6.9.2 Vibration. Perform this test utilizing the apparatus specified in Method 514.1 of MIL-STD-810 and in accordance with Part 1 of Procedure 1 therein, except that the vibratory conditions of 3.3.2.3 shall apply. Following vibration exposure, examine the assembly for structural failures, loose hardware, wire breakage, and abrasion. Perform the tests of 4.6.2 through 4.6.8 to determine compliance with 3.3.2.3

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TABLE VI
TRUTH TABLE

Item	Input to pins							Output to pins							
	10	23	24	35	37	33	14	29	30	6	17	16	19	18	32
1	1							0	1						
2	0							1	0						
3	1							0	1						
4		0	1							1					
5		1	0							0					
6		0	1							1					
7				1							0	1			
8				0							1	0			
9				1							0	1			
10					1								0	1	
11					0								1	0	
12					1								0	1	
13				1	1	0	1								1
14				1	1	1	1								1
15				1	1	1	0								0
16				1	1	1	1								0
17				0	1	1	1								0
18				0	0	1	1								1
19				1	1	1	1								1

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5. PREPARATION FOR DELIVERY

5.1 Packaging, packing and marking. Packaging, packing and marking shall be in accordance with Packaging Data Sheet 11738803. The level of protection shall be as specified in the procurement document.

6. NOTES

6.1 Intended use. The assembly is a module of the Laser Receiver-Transmitter RT-1021( )/VVG-l intended for use in the Fire Control Rangefinder AN/VVG-1( ). The assembly:

a. provides line drivers for the video, A-trigger signals and reset signals,

b. provides line receivers for test range signal, and

c. gene rates the malfunction 3 signal

6.2 Ordering data, Procurement documents should specify the following:

a. Title, number, and date of this specification,

b, Applicable packaging data sheet number (see 5.1),

c. Selection of applicable levels of preservation, packaging and packing,

d. Applicable stock number,

e. Requirement for First Article Submission.

Custodian:

Army - MU

Preparing Activity:

Army - MU

PROJECT NO. 1240-A315

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1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?
   A. GIVE PARAGRAPH NUMBER AND WORDING.

   B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

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